span

A span port transmits a copy of every Ethernet frame received by the bridge. The number of span ports configured on a bridge is unlimited, but if an interface is designated as a span port, it cannot also be used as a regular bridge port. This is most useful for snooping a bridged network passively on another host connected to one of the span ports of the bridge. For example, to send a copy of all frames out the interface named fxp4:

```
# ifconfig bridge0 span fxp4
```

sticky

If a bridge member interface is marked as sticky, dynamically learned address entries are treated as static entries in the forwarding cache. Sticky entries are never aged out of the cache or replaced, even if the address is seen on a different interface. This gives the benefit of static address entries without the need to pre-populate the forwarding table. Clients learned on a particular segment of the bridge cannot roam to another segment.

An example of using sticky addresses is to combine the bridge with VLANs in order to isolate customer networks without wasting IP address space. Consider that `CustomerA` is on `vlan100`, `CustomerB` is on `vlan101`, and the bridge has the address `192.168.0.1`:

```
# ifconfig bridge0 addm vlan100 sticky vlan100 addm vlan101 sticky vlan101
# ifconfig bridge0 inet 192.168.0.1/24
```

In this example, both clients see `192.168.0.1` as their default gateway. Since the bridge cache is sticky, one host cannot spoof the MAC address of the other customer in order to intercept their traffic.

Any communication between the VLANs can be blocked using a firewall or, as seen in this example, private interfaces:

```
# ifconfig bridge0 private vlan100 private vlan101
```

The customers are completely isolated from each other and the full `/24` address range can be allocated without subnetting.

The number of unique source MAC addresses behind an interface can be limited. Once the limit is reached, packets with unknown source addresses are dropped until an existing host cache entry expires or is removed.

The following example sets the maximum number of Ethernet devices for `CustomerA` on `vlan100` to 10:

```
# ifconfig bridge0 ifmaxaddr vlan100 10
```

Bridge interfaces also support monitor mode, where the packets are discarded after bpf(4)

processing and are not processed or forwarded further. This can be used to multiplex the input of two or more interfaces into a single bpf(4) stream. This is useful for reconstructing the traffic for network taps that transmit the RX/TX signals out through two separate interfaces. For example, to read the input from four network interfaces as one stream:

```
# ifconfig bridge0 addm fxp0 addm fxp1 addm fxp2 addm fxp3 monitor up
# tcpdump -i bridge0
```

34.8.4. SNMP Monitoring

The bridge interface and STP parameters can be monitored via bsnmpd(1) which is included in the FreeBSD base system. The exported bridge MIBs conform to IETF standards so any SNMP client or monitoring package can be used to retrieve the data.

To enable monitoring on the bridge, uncomment this line in /etc/snmpd.config by removing the beginning # symbol:

```
begemotSnmpdModulePath."bridge" = "/usr/lib/snmp_bridge.so"
```

Other configuration settings, such as community names and access lists, may need to be modified in this file. See bsnmpd(1) and snmp_bridge(3) for more information. Once these edits are saved, add this line to /etc/rc.conf:

```
bsnmpd_enable="YES"
```

Then, start bsnmpd(1):

```
# service bsnmpd start
```

The following examples use the Net-SNMP software (net-mgmt/net-snmp) to query a bridge from a client system. The net-mgmt/bsnmptools port can also be used. From the SNMP client which is running Net-SNMP, add the following lines to $HOME/.snmp/snmp.conf in order to import the bridge MIB definitions:

```
mibdirs +/usr/share/snmp/mibs
mibs +BRIDGE-MIB:RSTP-MIB:BEGEMOT-MIB:BEGEMOT-BRIDGE-MIB
```

To monitor a single bridge using the IETF BRIDGE-MIB (RFC4188):

```
% snmpwalk -v 2c -c public bridge1.example.com mib-2.dot1dBridge
BRIDGE-MIB::dot1dBaseBridgeAddress.0 = STRING: 66:fb:9b:6e:5c:44
BRIDGE-MIB::dot1dBaseNumPorts.0 = INTEGER: 1 ports
BRIDGE-MIB::dot1dStpTimeSinceTopologyChange.0 = Timeticks: (189959) 0:31:39.59 centi-
```

```
seconds
BRIDGE-MIB::dot1dStpTopChanges.0 = Counter32: 2
BRIDGE-MIB::dot1dStpDesignatedRoot.0 = Hex-STRING: 80 00 00 01 02 4B D4 50
...
BRIDGE-MIB::dot1dStpPortState.3 = INTEGER: forwarding(5)
BRIDGE-MIB::dot1dStpPortEnable.3 = INTEGER: enabled(1)
BRIDGE-MIB::dot1dStpPortPathCost.3 = INTEGER: 200000
BRIDGE-MIB::dot1dStpPortDesignatedRoot.3 = Hex-STRING: 80 00 00 01 02 4B D4 50
BRIDGE-MIB::dot1dStpPortDesignatedCost.3 = INTEGER: 0
BRIDGE-MIB::dot1dStpPortDesignatedBridge.3 = Hex-STRING: 80 00 00 01 02 4B D4 50
BRIDGE-MIB::dot1dStpPortDesignatedPort.3 = Hex-STRING: 03 80
BRIDGE-MIB::dot1dStpPortForwardTransitions.3 = Counter32: 1
RSTP-MIB::dot1dStpVersion.0 = INTEGER: rstp(2)
```

The `dot1dStpTopChanges.0` value is two, indicating that the STP bridge topology has changed twice. A topology change means that one or more links in the network have changed or failed and a new tree has been calculated. The `dot1dStpTimeSinceTopologyChange.0` value will show when this happened.

To monitor multiple bridge interfaces, the private BEGEMOT-BRIDGE-MIB can be used:

```
% snmpwalk -v 2c -c public bridge1.example.com
enterprises.fokus.begemot.begemotBridge
BEGEMOT-BRIDGE-MIB::begemotBridgeBaseName."bridge0" = STRING: bridge0
BEGEMOT-BRIDGE-MIB::begemotBridgeBaseName."bridge2" = STRING: bridge2
BEGEMOT-BRIDGE-MIB::begemotBridgeBaseAddress."bridge0" = STRING: e:ce:3b:5a:9e:13
BEGEMOT-BRIDGE-MIB::begemotBridgeBaseAddress."bridge2" = STRING: 12:5e:4d:74:d:fc
BEGEMOT-BRIDGE-MIB::begemotBridgeBaseNumPorts."bridge0" = INTEGER: 1
BEGEMOT-BRIDGE-MIB::begemotBridgeBaseNumPorts."bridge2" = INTEGER: 1
...
BEGEMOT-BRIDGE-MIB::begemotBridgeStpTimeSinceTopologyChange."bridge0" = Timeticks:
(116927) 0:19:29.27 centi-seconds
BEGEMOT-BRIDGE-MIB::begemotBridgeStpTimeSinceTopologyChange."bridge2" = Timeticks:
(82773) 0:13:47.73 centi-seconds
BEGEMOT-BRIDGE-MIB::begemotBridgeStpTopChanges."bridge0" = Counter32: 1
BEGEMOT-BRIDGE-MIB::begemotBridgeStpTopChanges."bridge2" = Counter32: 1
BEGEMOT-BRIDGE-MIB::begemotBridgeStpDesignatedRoot."bridge0" = Hex-STRING: 80 00 00 40
95 30 5E 31
BEGEMOT-BRIDGE-MIB::begemotBridgeStpDesignatedRoot."bridge2" = Hex-STRING: 80 00 00 50
8B B8 C6 A9
```

To change the bridge interface being monitored via the `mib-2.dot1dBridge` subtree:

```
% snmpset -v 2c -c private bridge1.example.com
BEGEMOT-BRIDGE-MIB::begemotBridgeDefaultBridgeIf.0 s bridge2
```

34.9. Link Aggregation and Failover

FreeBSD provides the lagg(4) interface which can be used to aggregate multiple network interfaces into one virtual interface in order to provide failover and link aggregation. Failover allows traffic to continue to flow as long as at least one aggregated network interface has an established link. Link aggregation works best on switches which support LACP, as this protocol distributes traffic bi-directionally while responding to the failure of individual links.

The aggregation protocols supported by the lagg interface determine which ports are used for outgoing traffic and whether or not a specific port accepts incoming traffic. The following protocols are supported by lagg(4):

failover

This mode sends and receives traffic only through the master port. If the master port becomes unavailable, the next active port is used. The first interface added to the virtual interface is the master port and all subsequently added interfaces are used as failover devices. If failover to a non-master port occurs, the original port becomes master once it becomes available again.

loadbalance

This provides a static setup and does not negotiate aggregation with the peer or exchange frames to monitor the link. If the switch supports LACP, that should be used instead.

lacp

The IEEE® 802.3ad Link Aggregation Control Protocol (LACP) negotiates a set of aggregable links with the peer into one or more Link Aggregated Groups (LAGs). Each LAG is composed of ports of the same speed, set to full-duplex operation, and traffic is balanced across the ports in the LAG with the greatest total speed. Typically, there is only one LAG which contains all the ports. In the event of changes in physical connectivity, LACP will quickly converge to a new configuration.

LACP balances outgoing traffic across the active ports based on hashed protocol header information and accepts incoming traffic from any active port. The hash includes the Ethernet source and destination address and, if available, the VLAN tag, and the IPv4 or IPv6 source and destination address.

roundrobin

This mode distributes outgoing traffic using a round-robin scheduler through all active ports and accepts incoming traffic from any active port. Since this mode violates Ethernet frame ordering, it should be used with caution.

broadcast

This mode sends outgoing traffic to all ports configured on the lagg interface, and receives frames on any port.

34.9.1. Configuration Examples

This section demonstrates how to configure a Cisco® switch and a FreeBSD system for LACP load balancing. It then shows how to configure two Ethernet interfaces in failover mode as well as how to configure failover mode between an Ethernet and a wireless interface.

Example 44. LACP Aggregation with a Cisco® Switch

This example connects two fxp(4) Ethernet interfaces on a FreeBSD machine to the first two Ethernet ports on a Cisco® switch as a single load balanced and fault tolerant link. More interfaces can be added to increase throughput and fault tolerance. Replace the names of the Cisco® ports, Ethernet devices, channel group number, and IP address shown in the example to match the local configuration.

Frame ordering is mandatory on Ethernet links and any traffic between two stations always flows over the same physical link, limiting the maximum speed to that of one interface. The transmit algorithm attempts to use as much information as it can to distinguish different traffic flows and balance the flows across the available interfaces.

On the Cisco® switch, add the *FastEthernet0/1* and *FastEthernet0/2* interfaces to channel group *1*:

```
interface FastEthernet0/1
 channel-group 1 mode active
 channel-protocol lacp
!
interface FastEthernet0/2
 channel-group 1 mode active
 channel-protocol lacp
```

On the FreeBSD system, create the lagg(4) interface using the physical interfaces *fxp0* and *fxp1* and bring the interfaces up with an IP address of *10.0.0.3/24*:

```
# ifconfig fxp0 up
# ifconfig fxp1 up
# ifconfig lagg0 create
# ifconfig lagg0 up laggproto lacp laggport fxp0 laggport fxp1 10.0.0.3/24
```

Next, verify the status of the virtual interface:

```
# ifconfig lagg0
lagg0: flags=8843<UP,BROADCAST,RUNNING,SIMPLEX,MULTICAST> metric 0 mtu 1500
        options=8<VLAN_MTU>
        ether 00:05:5d:71:8d:b8
        inet 10.0.0.3 netmask 0xffffff00 broadcast 10.0.0.255
        media: Ethernet autoselect
        status: active
        laggproto lacp
        laggport: fxp1 flags=1c<ACTIVE,COLLECTING,DISTRIBUTING>
        laggport: fxp0 flags=1c<ACTIVE,COLLECTING,DISTRIBUTING>
```

Ports marked as `ACTIVE` are part of the LAG that has been negotiated with the remote switch. Traffic will be transmitted and received through these active ports. Add `-v` to the above

command to view the LAG identifiers.

To see the port status on the Cisco® switch:

```
switch# show lacp neighbor
Flags:  S - Device is requesting Slow LACPDUs
        F - Device is requesting Fast LACPDUs
        A - Device is in Active mode       P - Device is in Passive mode

Channel group 1 neighbors

Partner's information:

                  LACP port                        Oper    Port     Port
Port      Flags   Priority  Dev ID         Age     Key     Number   State
Fa0/1     SA      32768     0005.5d71.8db8  29s    0x146   0x3      0x3D
Fa0/2     SA      32768     0005.5d71.8db8  29s    0x146   0x4      0x3D
```

For more detail, type `show lacp neighbor detail`.

To retain this configuration across reboots, add the following entries to /etc/rc.conf on the FreeBSD system:

```
ifconfig_fxp0="up"
ifconfig_fxp1="up"
cloned_interfaces="lagg0"
ifconfig_lagg0="laggproto lacp laggport fxp0 laggport fxp1 10.0.0.3/24"
```

Example 45. Failover Mode

Failover mode can be used to switch over to a secondary interface if the link is lost on the master interface. To configure failover, make sure that the underlying physical interfaces are up, then create the lagg(4) interface. In this example, *fxp0* is the master interface, *fxp1* is the secondary interface, and the virtual interface is assigned an IP address of *10.0.0.15/24*:

```
# ifconfig fxp0 up
# ifconfig fxp1 up
# ifconfig lagg0 create
# ifconfig lagg0 up laggproto failover laggport fxp0 laggport fxp1 10.0.0.15/24
```

The virtual interface should look something like this:

```
# ifconfig lagg0
lagg0: flags=8843<UP,BROADCAST,RUNNING,SIMPLEX,MULTICAST> metric 0 mtu 1500
        options=8<VLAN_MTU>
        ether 00:05:5d:71:8d:b8
```

```
        inet 10.0.0.15 netmask 0xffffff00 broadcast 10.0.0.255
        media: Ethernet autoselect
        status: active
        laggproto failover
        laggport: fxp1 flags=0<>
        laggport: fxp0 flags=5<MASTER,ACTIVE>
```

Traffic will be transmitted and received on *fxp0*. If the link is lost on *fxp0*, *fxp1* will become the active link. If the link is restored on the master interface, it will once again become the active link.

To retain this configuration across reboots, add the following entries to /etc/rc.conf:

```
ifconfig_fxp0="up"
ifconfig_fxp1="up"
cloned_interfaces="lagg0"
ifconfig_lagg0="laggproto failover laggport fxp0 laggport fxp1 10.0.0.15/24"
```

Example 46. Failover Mode Between Ethernet and Wireless Interfaces

For laptop users, it is usually desirable to configure the wireless device as a secondary which is only used when the Ethernet connection is not available. With lagg(4), it is possible to configure a failover which prefers the Ethernet connection for both performance and security reasons, while maintaining the ability to transfer data over the wireless connection.

This is achieved by overriding the Ethernet interface's MAC address with that of the wireless interface.

> In theory, either the Ethernet or wireless MAC address can be changed to match the other. However, some popular wireless interfaces lack support for overriding the MAC address. We therefore recommend overriding the Ethernet MAC address for this purpose.

> If the driver for the wireless interface is not loaded in the `GENERIC` or custom kernel, and the computer is running FreeBSD 12.1, load the corresponding .ko in /boot/loader.conf by adding `driver_load="YES"` to that file and rebooting. Another, better way is to load the driver in /etc/rc.conf by adding it to `kld_list` (see rc.conf(5) for details) in that file and rebooting. This is needed because otherwise the driver is not loaded yet at the time the lagg(4) interface is set up.

In this example, the Ethernet interface, *re0*, is the master and the wireless interface, *wlan0*, is the failover. The *wlan0* interface was created from the *ath0* physical wireless interface, and the Ethernet interface will be configured with the MAC address of the wireless interface. First, bring the wireless interface up (replacing *FR* with the local 2-letter country code), but do not

set an IP address. Replace *wlan0* to match the system's wireless interface name:

```
# ifconfig wlan0 create wlandev ath0 country FR ssid my_router up
```

Determine the MAC address of the wireless interface like this:

```
# ifconfig wlan0
wlan0: flags=8843<UP,BROADCAST,RUNNING,SIMPLEX,MULTICAST> metric 0 mtu 1500
    ether b8:ee:65:5b:32:59
    groups: wlan
    ssid Bbox-A3BD2403 channel 6 (2437 MHz 11g ht/20) bssid 00:37:b7:56:4b:60
    regdomain ETSI country FR indoor ecm authmode WPA2/802.11i privacy ON
    deftxkey UNDEF AES-CCM 2:128-bit txpower 30 bmiss 7 scanvalid 60
    protmode CTS ampdulimit 64k ampdudensity 8 shortgi -stbctx stbcrx
    -ldpc wme burst roaming MANUAL
    media: IEEE 802.11 Wireless Ethernet MCS mode 11ng
    status: associated
    nd6 options=29<PERFORMNUD,IFDISABLED,AUTO_LINKLOCAL>
```

The `ether` line will contain the MAC address of the specified interface. Now, change the MAC address of the Ethernet interface to match:

```
# ifconfig re0 ether b8:ee:65:5b:32:59
```

Make sure the *re0* interface is up, then create the lagg(4) interface with *re0* as master with failover to *wlan0*:

```
# ifconfig re0 up
# ifconfig lagg0 create
# ifconfig lagg0 up laggproto failover laggport re0 laggport wlan0
```

The virtual interface should look something like this:

```
# ifconfig lagg0
lagg0: flags=8843<UP,BROADCAST,RUNNING,SIMPLEX,MULTICAST> metric 0 mtu 1500
        options=8<VLAN_MTU>
        ether b8:ee:65:5b:32:59
        laggproto failover lagghash l2,l3,l4
        laggport: re0 flags=5<MASTER,ACTIVE>
        laggport: wlan0 flags=0<>
        groups: lagg
        media: Ethernet autoselect
        status: active
```

Then, start the DHCP client to obtain an IP address:

```
# dhclient lagg0
```

To retain this configuration across reboots, add the following entries to /etc/rc.conf:

```
ifconfig_re0="ether b8:ee:65:5b:32:59"
wlans_ath0="wlan0"
ifconfig_wlan0="WPA"
create_args_wlan0="country FR"
cloned_interfaces="lagg0"
ifconfig_lagg0="up laggproto failover laggport re0 laggport wlan0 DHCP"
```

34.10. Diskless Operation with PXE

The Intel® Preboot eXecution Environment (PXE) allows an operating system to boot over the network. For example, a FreeBSD system can boot over the network and operate without a local disk, using file systems mounted from an NFS server. PXE support is usually available in the BIOS. To use PXE when the machine starts, select the `Boot from network` option in the BIOS setup or type a function key during system initialization.

In order to provide the files needed for an operating system to boot over the network, a PXE setup also requires properly configured DHCP, TFTP, and NFS servers, where:

- Initial parameters, such as an IP address, executable boot filename and location, server name, and root path are obtained from the DHCP server.
- The operating system loader file is booted using TFTP.
- The file systems are loaded using NFS.

When a computer PXE boots, it receives information over DHCP about where to obtain the initial boot loader file. After the host computer receives this information, it downloads the boot loader via TFTP and then executes the boot loader. In FreeBSD, the boot loader file is /boot/pxeboot. After /boot/pxeboot executes, the FreeBSD kernel is loaded and the rest of the FreeBSD bootup sequence proceeds, as described in The FreeBSD Booting Process.



For UEFI PXE based boot, the actual boot loader file to use is /boot/loader.efi. See the below section Debugging PXE Problems on how to use /boot/loader.efi.

This section describes how to configure these services on a FreeBSD system so that other systems can PXE boot into FreeBSD. Refer to diskless(8) for more information.



As described, the system providing these services is insecure. It should live in a protected area of a network and be untrusted by other hosts.

34.10.1. Setting Up the PXE Environment

The steps shown in this section configure the built-in NFS and TFTP servers. The next section demonstrates how to install and configure the DHCP server. In this example, the directory which will contain the files used by PXE users is /b/tftpboot/FreeBSD/install. It is important that this directory exists and that the same directory name is set in both /etc/inetd.conf and /usr/local/etc/dhcpd.conf.



The command examples below assume use of the sh(1) shell. csh(1) and tcsh(1) users will need to start a sh(1) shell or adapt the commands to csh(1) syntax.

1. Create the root directory which will contain a FreeBSD installation to be NFS mounted:

```
# export NFSROOTDIR=/b/tftpboot/FreeBSD/install
# mkdir -p ${NFSROOTDIR}
```

2. Enable the NFS server by adding this line to /etc/rc.conf:

```
nfs_server_enable="YES"
```

3. Export the diskless root directory via NFS by adding the following to /etc/exports:

```
/b -ro -alldirs -maproot=root
```

4. Start the NFS server:

```
# service nfsd start
```

5. Enable inetd(8) by adding the following line to /etc/rc.conf:

```
inetd_enable="YES"
```

6. Uncomment the following line in /etc/inetd.conf by making sure it does not start with a # symbol:

```
tftp dgram udp wait root /usr/libexec/tftpd tftpd blocksize 1468 -l -s /b/tftpboot
```



The specified tftp blocksize, e.g. 1468 bytes, replaces the default size 512 bytes. Some PXE versions require the TCP version of TFTP. In this case, uncomment the second `tftp` line which contains `stream tcp`.

7. Start inetd(8):

```
# service inetd start
```

8. Install the base system into ${NFSROOTDIR}, either by decompressing the official archives or by rebuilding the FreeBSD kernel and userland (refer to "Updating FreeBSD from Source" for more detailed instructions, but do not forget to add `DESTDIR=${NFSROOTDIR}` when running the `make installkernel` and `make installworld` commands.
9. Test that the TFTP server works and can download the boot loader which will be obtained via PXE:

```
# tftp localhost
tftp> get FreeBSD/install/boot/pxeboot
Received 264951 bytes in 0.1 seconds
```

10. Edit ${NFSROOTDIR}/etc/fstab and create an entry to mount the root file system over NFS:

```
# Device                                    Mountpoint    FSType   Options
Dump Pass
myhost.example.com:/b/tftpboot/FreeBSD/install      /         nfs      ro        0
0
```

    Replace *myhost.example.com* with the hostname or IP address of the NFS server. In this example, the root file system is mounted read-only in order to prevent NFS clients from potentially deleting the contents of the root file system.

11. Set the root password in the PXE environment for client machines which are PXE booting :

```
# chroot ${NFSROOTDIR}
# passwd
```

12. If needed, enable ssh(1) root logins for client machines which are PXE booting by editing ${NFSROOTDIR}/etc/ssh/sshd_config and enabling `PermitRootLogin`. This option is documented in sshd_config(5).
13. Perform any other needed customizations of the PXE environment in ${NFSROOTDIR}. These customizations could include things like installing packages or editing the password file with vipw(8).

When booting from an NFS root volume, /etc/rc detects the NFS boot and runs /etc/rc.initdiskless. In this case, /etc and /var need to be memory backed file systems so that these directories are writable but the NFS root directory is read-only:

```
# chroot ${NFSROOTDIR}
# mkdir -p conf/base
# tar -c -v -f conf/base/etc.cpio.gz --format cpio --gzip etc
```

```
# tar -c -v -f conf/base/var.cpio.gz --format cpio --gzip var
```

When the system boots, memory file systems for /etc and /var will be created and mounted and the contents of the cpio.gz files will be copied into them. By default, these file systems have a maximum capacity of 5 megabytes. If the archives do not fit, which is usually the case for /var when binary packages have been installed, request a larger size by putting the number of 512 byte sectors needed (e.g., 5 megabytes is 10240 sectors) in ${NFSROOTDIR}/conf/base/etc/md_size and ${NFSROOTDIR}/conf/base/var/md_size files for /etc and /var file systems respectively.

34.10.2. Configuring the DHCP Server

The DHCP server does not need to be the same machine as the TFTP and NFS server, but it needs to be accessible in the network.

DHCP is not part of the FreeBSD base system but can be installed using the net/isc-dhcp44-server port or package.

Once installed, edit the configuration file, /usr/local/etc/dhcpd.conf. Configure the `next-server`, `filename`, and `root-path` settings as seen in this example:

```
subnet 192.168.0.0 netmask 255.255.255.0 {
   range 192.168.0.2 192.168.0.3 ;
   option subnet-mask 255.255.255.0 ;
   option routers 192.168.0.1 ;
   option broadcast-address 192.168.0.255 ;
   option domain-name-servers 192.168.35.35, 192.168.35.36 ;
   option domain-name "example.com";

   # IP address of TFTP server
   next-server 192.168.0.1 ;

   # path of boot loader obtained via tftp
   filename "FreeBSD/install/boot/pxeboot" ;

   # pxeboot boot loader will try to NFS mount this directory for root FS
   option root-path "192.168.0.1:/b/tftpboot/FreeBSD/install/" ;

}
```

The `next-server` directive is used to specify the IP address of the TFTP server.

The `filename` directive defines the path to /boot/pxeboot. A relative filename is used, meaning that /b/tftpboot is not included in the path.

The `root-path` option defines the path to the NFS root file system.

Once the edits are saved, enable DHCP at boot time by adding the following line to /etc/rc.conf:

```
dhcpd_enable="YES"
```

Then start the DHCP service:

```
# service isc-dhcpd start
```

34.10.3. Debugging PXE Problems

Once all of the services are configured and started, PXE clients should be able to automatically load FreeBSD over the network. If a particular client is unable to connect, when that client machine boots up, enter the BIOS configuration menu and confirm that it is set to boot from the network.

This section describes some troubleshooting tips for isolating the source of the configuration problem should no clients be able to PXE boot.

1. Use the net/wireshark package or port to debug the network traffic involved during the PXE booting process, which is illustrated in the diagram below.

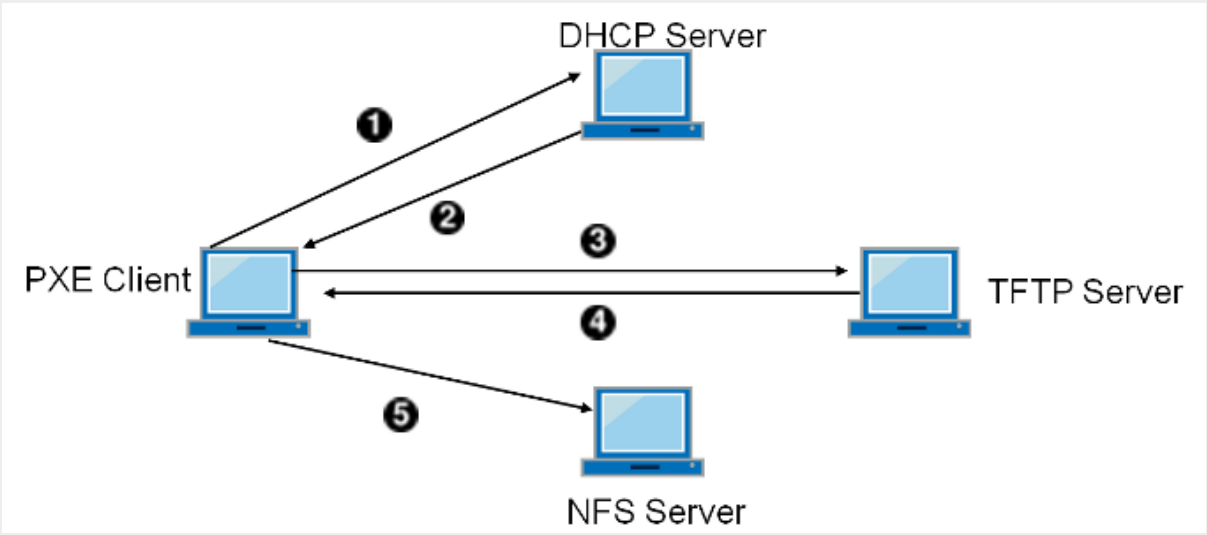


Figure 78. PXE Booting Process with NFS Root Mount

   1. Client broadcasts a DHCPDISCOVER message.
   2. The DHCP server responds with the IP address, next-server, filename, and root-path values.
   3. The client sends a TFTP request to next-server, asking to retrieve filename.
   4. The TFTP server responds and sends filename to client.
   5. The client executes filename, which is pxeboot(8), which then loads the kernel. When the kernel executes, the root file system specified by root-path is mounted over NFS.
2. On the TFTP server, read /var/log/xferlog to ensure that pxeboot is being retrieved from the correct location. To test this example configuration:

```
# tftp 192.168.0.1
```

```
tftp> get FreeBSD/install/boot/pxeboot
Received 264951 bytes in 0.1 seconds
```

The BUGS sections in tftpd(8) and tftp(1) document some limitations with TFTP.

3. Make sure that the root file system can be mounted via NFS. To test this example configuration:

```
# mount -t nfs 192.168.0.1:/b/tftpboot/FreeBSD/install /mnt
```

4. For UEFI PXE based booting, replace the boot/pxeboot file with the boot/loader.efi file:

```
# chroot ${NFSROOTDIR}
# mv boot/pxeboot boot/pxeboot.original
# cp boot/loader.efi boot/pxeboot
```

34.11. Common Address Redundancy Protocol (CARP)

The Common Address Redundancy Protocol (CARP) allows multiple hosts to share the same IP address and Virtual Host ID (VHID) in order to provide *high availability* for one or more services. This means that one or more hosts can fail, and the other hosts will transparently take over so that users do not see a service failure.

In addition to the shared IP address, each host has its own IP address for management and configuration. All of the machines that share an IP address have the same VHID. The VHID for each virtual IP address must be unique across the broadcast domain of the network interface.

High availability using CARP is built into FreeBSD, though the steps to configure it vary slightly depending upon the FreeBSD version. This section provides the same example configuration for versions before and equal to or after FreeBSD 10.

This example configures failover support with three hosts, all with unique IP addresses, but providing the same web content. It has two different masters named `hosta.example.org` and `hostb.example.org`, with a shared backup named `hostc.example.org`.

These machines are load balanced with a Round Robin DNS configuration. The master and backup machines are configured identically except for their hostnames and management IP addresses. These servers must have the same configuration and run the same services. When the failover occurs, requests to the service on the shared IP address can only be answered correctly if the backup server has access to the same content. The backup machine has two additional CARP interfaces, one for each of the master content server's IP addresses. When a failure occurs, the backup server will pick up the failed master machine's IP address.

34.11.1. Using CARP

Enable boot-time support for CARP by adding an entry for the carp.ko kernel module in

/boot/loader.conf:

```
carp_load="YES"
```

To load the module now without rebooting:

```
# kldload carp
```

For users who prefer to use a custom kernel, include the following line in the custom kernel configuration file and compile the kernel as described in Configuring the FreeBSD Kernel:

```
device  carp
```

The hostname, management IP address and subnet mask, shared IP address, and VHID are all set by adding entries to /etc/rc.conf. This example is for `hosta.example.org`:

```
hostname="hosta.example.org"
ifconfig_em0="inet 192.168.1.3 netmask 255.255.255.0"
ifconfig_em0_alias0="inet vhid 1 pass testpass alias 192.168.1.50/32"
```

The next set of entries are for `hostb.example.org`. Since it represents a second master, it uses a different shared IP address and VHID. However, the passwords specified with `pass` must be identical as CARP will only listen to and accept advertisements from machines with the correct password.

```
hostname="hostb.example.org"
ifconfig_em0="inet 192.168.1.4 netmask 255.255.255.0"
ifconfig_em0_alias0="inet vhid 2 pass testpass alias 192.168.1.51/32"
```

The third machine, `hostc.example.org`, is configured to handle failover from either master. This machine is configured with two CARPVHIDs, one to handle the virtual IP address for each of the master hosts. The CARP advertising skew, `advskew`, is set to ensure that the backup host advertises later than the master, since `advskew` controls the order of precedence when there are multiple backup servers.

```
hostname="hostc.example.org"
ifconfig_em0="inet 192.168.1.5 netmask 255.255.255.0"
ifconfig_em0_alias0="inet vhid 1 advskew 100 pass testpass alias 192.168.1.50/32"
ifconfig_em0_alias1="inet vhid 2 advskew 100 pass testpass alias 192.168.1.51/32"
```

Having two CARPVHIDs configured means that `hostc.example.org` will notice if either of the master servers becomes unavailable. If a master fails to advertise before the backup server, the backup server will pick up the shared IP address until the master becomes available again.



If the original master server becomes available again, `hostc.example.org` will not release the virtual IP address back to it automatically. For this to happen, preemption has to be enabled. The feature is disabled by default, it is controlled via the sysctl(8) variable `net.inet.carp.preempt`. The administrator can force the backup server to return the IP address to the master:

```
# ifconfig em0 vhid 1 state backup
```

Once the configuration is complete, either restart networking or reboot each system. High availability is now enabled.

CARP functionality can be controlled via several sysctl(8) variables documented in the carp(4) manual pages. Other actions can be triggered from CARP events by using devd(8).

34.12. VLANs

VLANs are a way of virtually dividing up a network into many different subnetworks, also referred to as segmenting. Each segment will have its own broadcast domain and be isolated from other VLANs.

On FreeBSD, VLANs must be supported by the network card driver. To see which drivers support vlans, refer to the vlan(4) manual page.

When configuring a VLAN, a couple pieces of information must be known. First, which network interface? Second, what is the VLAN tag?

To configure VLANs at run time, with a NIC of `em0` and a VLAN tag of `5` the command would look like this:

```
# ifconfig em0.5 create vlan 5 vlandev em0 inet 192.168.20.20/24
```



See how the interface name includes the NIC driver name and the VLAN tag, separated by a period? This is a best practice to make maintaining the VLAN configuration easy when many VLANs are present on a machine.



When defining VLANs, ensure that the parent network interface is also configured and enabled. The minimum configuration for the above example would be:

```
# ifconfig em0 up
```

To configure VLANs at boot time, /etc/rc.conf must be updated. To duplicate the configuration above, the following will need to be added:

```
vlans_em0="5"
```

```
ifconfig_em0_5="inet 192.168.20.20/24"
```

Additional VLANs may be added, by simply adding the tag to the `vlans_em0` field and adding an additional line configuring the network on that VLAN tag's interface.



When defining VLANs in /etc/rc.conf, make sure that the parent network interface is configured and enabled as well. The minimum configuration for the above example would be:

```
ifconfig_em0="up"
```

It is useful to assign a symbolic name to an interface so that when the associated hardware is changed, only a few configuration variables need to be updated. For example, security cameras need to be run over VLAN 1 on `em0`. Later, if the `em0` card is replaced with a card that uses the ixgb(4) driver, all references to `em0.1` will not have to change to `ixgb0.1`.

To configure VLAN `5`, on the NIC `em0`, assign the interface name `cameras`, and assign the interface an IP address of `192.168.20.20` with a `24`-bit prefix, use this command:

```
# ifconfig em0.5 create vlan 5 vlandev em0 name cameras inet 192.168.20.20/24
```

For an interface named `video`, use the following:

```
# ifconfig video.5 create vlan 5 vlandev video name cameras inet 192.168.20.20/24
```

To apply the changes at boot time, add the following lines to /etc/rc.conf:

```
vlans_video="cameras"
create_args_cameras="vlan 5"
ifconfig_cameras="inet 192.168.20.20/24"
```

Part V: Appendices

Appendix A: Obtaining FreeBSD

A.1. Mirrors

The official mirrors of the FreeBSD project are made up of many machines operated by the project cluster administrators and behind GeoDNS to direct users to the closest available mirror. Current locations are Australia, Brazil, Japan (two sites), Malaysia, South Africa, Sweden, Taiwan, United States of America (California - two sites, Illinois - two sites, New Jersey, and Washington).

Official mirrors service:

Service Name	Protocols	More information
cgit.FreeBSD.org	https	Web interface for the FreeBSD Git repositories.
docs.FreeBSD.org	https	FreeBSD Documentation Portal.
download.FreeBSD.org	https ftp	Same content as `ftp.FreeBSD.org`, `ftp` is a legacy name; `download.FreeBSD.org` is recommended.
git.FreeBSD.org	git over `https` and `ssh`	More details on using git section.
pkg.FreeBSD.org	pkg(8) over `http` and `https`	Official FreeBSD package repositories used by the pkg(8) program.
vuxml.FreeBSD.org / www.VuXML.org	https	FreeBSD Project VuXML web page. `pkg audit` fetches the list of vulnerabilities from this service.
www.FreeBSD.org	https	FreeBSD Website.

All official mirrors support IPv4 and IPv6.

http://ftp-archive.FreeBSD.org is not in the GeoDNS Infrastructure, hosted in only one location (US).

The project is looking for new locations; those willing to sponsor, please reach out to the Cluster Administrators team for more information.

Fastly CDN is used to serve **download.FreeBSD.org** and **pkg.FreeBSD.org**, providing acceleration, caching, and global bandwidth distribution for FreeBSD images and packages.

Mirror list maintained by the community and other companies:

Country	Hostname	Protocols
Australia ✉	ftp.au.FreeBSD.org	http http_v6 rsync rsync_v6
	ftp3.au.FreeBSD.org	http ftp rsync
Brazil ✉	ftp2.br.FreeBSD.org	http rsync rsync_v6

Country	Hostname	Protocols
	ftp3.br.FreeBSD.org	http ftp rsync
Bulgaria	ftp.bg.FreeBSD.org	ftp ftp_v6 rsync rsync_v6
Czech Republic	ftp.cz.FreeBSD.org	http http_v6 rsync rsync_v6
Denmark	ftp.dk.FreeBSD.org	http http_v6 ftp ftp_v6 rsync rsync_v6
Finland	ftp.fi.FreeBSD.org	ftp
France	ftp.fr.FreeBSD.org	http http_v6 ftp ftp_v6 rsync rsync_v6
	ftp3.fr.FreeBSD.org	ftp
	ftp6.fr.FreeBSD.org	http ftp rsync
Germany	ftp.de.FreeBSD.org	http http_v6 https https_v6 ftp ftp_v6 rsync rsync_v6
	ftp1.de.FreeBSD.org	http http_v6 ftp ftp_v6 rsync rsync_v6
	ftp2.de.FreeBSD.org	http http_v6 ftp ftp_v6 rsync rsync_v6
	ftp5.de.FreeBSD.org	ftp ftp_v6
	ftp7.de.FreeBSD.org	http http_v6 ftp ftp_v6
Greece	ftp.gr.FreeBSD.org	http http_v6 ftp ftp_v6
	ftp2.gr.FreeBSD.org	http http_v6 ftp ftp_v6 rsync
Japan	ftp.jp.FreeBSD.org	http http_v6 ftp ftp_v6 rsync rsync_v6
	ftp2.jp.FreeBSD.org	ftp rsync rsync_v6
	ftp3.jp.FreeBSD.org	http rsync
	ftp4.jp.FreeBSD.org	ftp
	ftp6.jp.FreeBSD.org	http http_v6 ftp ftp_v6 rsync rsync_v6
Kazakhstan	mirror.ps.kz	http ftp
	mirror.neolabs.kz	http ftp
Korea	ftp.kr.FreeBSD.org	http https ftp rsync
	ftp2.kr.FreeBSD.org	rsync

Country	Hostname	Protocols
Latvia	ftp.lv.FreeBSD.org	http ftp
Netherlands	ftp.nl.FreeBSD.org	http http_v6 ftp ftp_v6 rsync rsync_v6
	ftp2.nl.FreeBSD.org	http ftp rsync
	mirror.nl.altushost.com	https
New Zealand	ftp.nz.FreeBSD.org	http ftp
Norway	ftp.no.FreeBSD.org	ftp ftp_v6 rsync rsync_v6
Poland	ftp.pl.FreeBSD.org	http http_v6 ftp rsync rsync_v6
Russia	ftp.ru.FreeBSD.org	http http_v6 ftp ftp_v6 rsync rsync_v6
	ftp2.ru.FreeBSD.org	https ftp rsync
Slovenia	ftp.si.FreeBSD.org	http http_v6 ftp ftp_v6
South Africa	ftp2.za.FreeBSD.org	http http_v6 ftp_v6
	ftp4.za.FreeBSD.org	http ftp rsync
Sweden	ftp.se.FreeBSD.org	http http_v6 ftp ftp_v6 rsync rsync_v6
	mirror.se.altushost.com	https
Taiwan	ftp4.tw.FreeBSD.org	https ftp rsync
	ftp5.tw.FreeBSD.org	http ftp
Ukraine	ftp.ua.FreeBSD.org	http ftp ftp_v6 rsync rsync_v6
United Kingdom	ftp.uk.FreeBSD.org	http http_v6 ftp ftp_v6 rsync rsync_v6
	ftp2.uk.FreeBSD.org	http http_v6 https https_v6 ftp ftp_v6
United States of America	ftp11.FreeBSD.org	http http_v6 ftp ftp_v6 rsync rsync_v6
	ftp14.FreeBSD.org	ftp rsync (Former official tier 1)
	ftp5.FreeBSD.org	http http_v6 ftp ftp_v6

The current list of protocols supported by the community mirrors was last updated on 2022-01-31, and it's not guaranteed.

A.2. Using Git

A.2.1. Introduction

As of December 2020, FreeBSD uses git as the primary version control system for storing all of FreeBSD's base source code and documentation. As of April 2021, FreeBSD uses git as the only version control system for storing all of FreeBSD's Ports Collection.



Git is generally a developer tool. Users may prefer to use `freebsd-update` ("FreeBSD Update") to update the FreeBSD base system.

This section demonstrates how to install Git on a FreeBSD system and use it to create a local copy of a FreeBSD source code repository.

A.2.2. Installation

Git can be installed from the Ports Collection, or as a package:

```
# pkg install git
```

A.2.3. Running Git

To fetch a clean copy of the sources into a local directory, use `git clone`. This directory of files is called the *working tree*.

Git uses URLs to designate a repository. There are three different repositories, `src` for the FreeBSD system source code, `doc` for documentation, and `ports` for the FreeBSD Ports Collection. All three are reachable over two different protocols: HTTPS and SSH. For example, the URL `https://git.FreeBSD.org/src.git` specifies the main branch of the `src` repository, using the `https` protocol.

Table 51. FreeBSD Git Repository URL Table

Item	Git URL
Read-only src repo via HTTPS	`https://git.FreeBSD.org/src.git`
Read-only src repo via anon-ssh	`ssh://anongit@git.FreeBSD.org/src.git`
Read-only doc repo via HTTPS	`https://git.FreeBSD.org/doc.git`
Read-only doc repo via anon-ssh	`ssh://anongit@git.FreeBSD.org/doc.git`
Read-only ports repo via HTTPS	`https://git.FreeBSD.org/ports.git`
Read-only ports repo via anon-ssh	`ssh://anongit@git.FreeBSD.org/ports.git`

External mirrors maintained by project members are also available; please refer to the External mirrors section.

To clone a copy of the FreeBSD system source code repository:

```
# git clone -o freebsd https://git.FreeBSD.org/src.git /usr/src
```

The `-o freebsd` option specifies the origin; by convention in the FreeBSD documentation, the origin is assumed to be `freebsd`. Because the initial checkout must download the full branch of the remote repository, it can take a while. Please be patient.

Initially, the working tree contains source code for the `main` branch, which corresponds to CURRENT. To switch to 13-STABLE instead:

```
# cd /usr/src
# git checkout stable/13
```

The working tree can be updated with `git pull`. To update /usr/src created in the example above, use:

```
# cd /usr/src
# git pull --rebase
```

The update is much quicker than a checkout, only transferring files that have changed.

A.2.4. Web-based repository browser

The FreeBSD project uses cgit as the web-based repository browser: https://cgit.FreeBSD.org/.

A.2.5. For Developers

For information about write access to repositories see the Committer's Guide.

A.2.6. External mirrors

Those mirrors are not hosted in FreeBSD.org but still maintained by the project members. Users and developers are welcome to pull or browse repositories on those mirrors. Pull requests for the `doc` and `src` GitHub repositories are being accepted; otherwise, the project workflow with those mirrors is still under discussion.

Codeberg

- doc: https://codeberg.org/FreeBSD/freebsd-doc
- ports: https://codeberg.org/FreeBSD/freebsd-ports
- src: https://codeberg.org/FreeBSD/freebsd-src

GitHub

- doc: https://github.com/freebsd/freebsd-doc
- ports: https://github.com/freebsd/freebsd-ports
- src: https://github.com/freebsd/freebsd-src

GitLab

- doc: https://gitlab.com/FreeBSD/freebsd-doc
- ports: https://gitlab.com/FreeBSD/freebsd-ports
- src: https://gitlab.com/FreeBSD/freebsd-src

A.2.7. Mailing lists

The main mailing list for general usage and questions about git in the FreeBSD project is freebsd-git. For more details, including commit messages lists, see the Mailing Lists chapter.

A.2.8. SSH host keys

- gitrepo.FreeBSD.org host key fingerprints:
  - ECDSA key fingerprint is `SHA256:seWO5D27ySURcx4bknTNKlC1mgai0whP443PAKEvvZA`
  - ED25519 key fingerprint is `SHA256:lNR6i4BEOaaUhmDHBA1WJsO7H3KtvjE2r5q4sOxtIWo`
  - RSA key fingerprint is `SHA256:f453CUEFXEJAXlKeEHV+ajJfeEfx9MdKQUD7lIscnQI`
- git.FreeBSD.org host key fingerprints:
  - ECDSA key fingerprint is `SHA256:/UlirUAsGiitupxmtsn7f9b7zCWd0vCs4Yo/tpVWP9w`
  - ED25519 key fingerprint is `SHA256:y1ljKrKMD3lDObRUG3xJ9gXwEIuqnh306tSyFd1tuZE`
  - RSA key fingerprint is `SHA256:jBe6FQGoH4HjvrIVM23dcnLZk9kmpdezR/CvQzm7rJM`

These are also published as SSHFP records in DNS.

Appendix B: Bibliography

While manual pages provide a definitive reference for individual pieces of the FreeBSD operating system, they seldom illustrate how to put the pieces together to make the whole operating system run smoothly. For this, there is no substitute for a good book or users' manual on UNIX® system administration.

B.1. FreeBSD Bibliography

- **Absolute FreeBSD: The Complete Guide To FreeBSD**, Third Edition, published by No Starch Press, 2018. ISBN: 978-1593278922
- **FreeBSD Mastery: Storage Essentials**, published by Tilted Windmill Press, 2014. ISBN: 978-1642350098
- **FreeBSD Mastery: Specialty Filesystems**, published by Tilted Windmill Press, 2015. ISBN: 978-1642350111
- **FreeBSD Mastery: ZFS**, published by Tilted Windmill Press, 2015. ISBN: 978-1642350005
- **FreeBSD Mastery: Advanced ZFS**, published by Tilted Windmill Press, 2016. ISBN: 978-0692688687
- **FreeBSD Mastery: Jails**, published by Tilted Windmill Press, 2019. ISBN: 978-1642350241
- **FreeBSD Device Drivers: A Guide for the Intrepid**, published by No Starch Press, 2012. ISBN: 978-1593272043
- **The Design And Implementation Of The Freebsd Operating System**, Second Edition, published by Pearson Education, Inc., 2014. ISBN: 978-0321968975
- **UNIX and Linux System Administration Handbook**, Fifth Edition, published by Pearson Education, Inc., 2017. ISBN: 978-0134277554
- **Designing BSD Rootkits**, published by No Starch Press, 2007. ISBN: 978-1593271428
- **FreeBSD Jails using VNETs**, published in gumroad

B.2. Security Reference

- **The Book of PF: A No-Nonsense Guide to the OpenBSD Firewall**, Third Edition, published by No Starch Press, 2014. ISBN: 978-1593275891
- **SSH Mastery: OpenSSH, PuTTY, Tunnels, and Keys**, Second Edition, 2018. ISBN: 978-1642350029

B.3. UNIX® History

- Lion, John *Lion's Commentary on UNIX, 6th Ed. With Source Code*. ITP Media Group, 1996. ISBN 1573980137
- Raymond, Eric S. *The New Hacker's Dictionary, 3rd edition*. MIT Press, 1996. ISBN 0-262-68092-0. Also known as the Jargon File

- Salus, Peter H. *A quarter century of UNIX*. Addison-Wesley Publishing Company, Inc., 1994. ISBN 0-201-54777-5
- Simon Garfinkel, Daniel Weise, Steven Strassmann. *The UNIX-HATERS Handbook*. IDG Books Worldwide, Inc., 1994. ISBN 1-56884-203-1. Out of print, but available online.
- Don Libes, Sandy Ressler *Life with UNIX* - special edition. Prentice-Hall, Inc., 1989. ISBN 0-13-536657-7
- *The BSD family tree*. https://cgit.freebsd.org/src/tree/share/misc/bsd-family-tree or /usr/share/misc/bsd-family-tree on a FreeBSD machine.
- *Networked Computer Science Technical Reports Library*.
- *Old BSD releases from the Computer Systems Research group (CSRG)*. http://www.mckusick.com/csrg/: The 4CD set covers all BSD versions from 1BSD to 4.4BSD and 4.4BSD-Lite2 (but not 2.11BSD, unfortunately). The last disk also holds the final sources plus the SCCS files.
- Kernighan, Brian *Unix: A History and a Memoir*. Kindle Direct Publishing, 2020. ISBN 978-169597855-3

B.4. Periodicals, Journals, and Magazines

- Admin Magazin (in German), published by Medialinx AG. ISSN: 2190-1066
- BSD Now - Video Podcast, published by Jupiter Broadcasting LLC
- FreeBSD Journal, published by S&W Publishing, sponsored by The FreeBSD Foundation. ISBN: 978-0-615-88479-0

Appendix C: Resources on the Internet

Development of FreeBSD is too rapid for print media to be practical for keeping people informed. For awareness of developments: electronic alternatives to print are best.

The FreeBSD user community provides much technical support — with forums, chat and email amongst the most popular and effective means of communication.

The most important points of contact are outlined below. The Community wiki area may be more up-to-date.

Please make the FreeBSD documentation project mailing list aware of any resource that is either redundant, or not yet listed below.

C.1. Websites

- The FreeBSD Forums provide a web based discussion forum for FreeBSD questions and technical discussion.
- The FreeBSD Wiki provides various bits of information that hadn't yet made it into the Handbook.
- The Documentation Portal offers much more than the FreeBSD Handbook alone; there are more than forty books and articles.
- The FreeBSD Journal is a free, professionally-edited, bi-monthly technical magazine released by The FreeBSD Foundation.
- The BSDConferences YouTube Channel provides a collection of high quality videos from BSD conferences around the world. This is a great way to watch key developers give presentations about new work in FreeBSD.
- FreeBSD Status Reports are released every three months and track progress of FreeBSD development.
- There's a FreeBSD-focused Reddit group at r/freebsd.
- Super User and Server Fault, the Stack Exchange services for system administrators.
- FreeBSD Discord server, a communications and community-building service, where FreeBSD community members can socialise, obtain support or support others, learn, contribute, collaborate, and stay up to date with all things FreeBSD-related.
- IRC channels, a widely implemented, technically mature, open standard text chat.

C.2. Mailing Lists

The mailing lists are the most direct way of addressing questions or opening a technical discussion to a concentrated FreeBSD audience. There are a wide variety of lists on a number of different FreeBSD topics. Sending questions to the most appropriate mailing list will invariably assure a faster and more accurate response.

Technical list threads should remain technical.

All users and developers of FreeBSD should subscribe to the FreeBSD announcements mailing list.



To test FreeBSD mailing list capabilities, aim for the FreeBSD test mailing list. Please do not send test messages to any other list.

When in doubt about what list to post a question to, see How to get best results from the FreeBSD-questions mailing list.

Before posting to any list, please:

- learn about how to best use the mailing lists, such as how to help avoid frequently-repeated discussions, by reading the Mailing List Frequently Asked Questions (FAQ) document
- search the archives, to tell whether someone else has already posted the same.

Archive search interfaces include:

- https://lists.freebsd.org/search (FreeBSD, experimental)
- https://www.freebsd.org/search/ (DuckDuckGo)

Note that this also means that messages sent to FreeBSD mailing lists are archived in perpetuity. When protecting privacy is a concern, consider using a disposable secondary email address and posting only public information.

FreeBSD-provided archives:

- do not present links as links
- do not present inline images
- do not present HTML content of HTML messages.

The FreeBSD public mailing lists can be consulted here.

C.2.1. How to Subscribe or Unsubscribe

At https://lists.freebsd.org, click the name of a list to reveal its options.

To post, after subscribing, send mail to `listname@FreeBSD.org`. The message will be redistributed to list members.

C.2.2. Lists Basic Rules

All FreeBSD mailing lists have certain basic rules which must be adhered to by anyone using them. Failure to comply with these guidelines will result in two (2) written warnings from the FreeBSD Postmaster postmaster@FreeBSD.org, after which, on a third offense, the poster will removed from all FreeBSD mailing lists and filtered from further posting to them. We regret that such rules and measures are necessary at all, but today's Internet is a pretty harsh environment, it would seem, and many fail to appreciate just how fragile some of its mechanisms are.

Rules of the road:

- The topic of any posting should adhere to the basic description of the list it is posted to. If the list is about technical issues, the posting should contain technical discussion. Ongoing irrelevant chatter or flaming only detracts from the value of the mailing list for everyone on it and will not be tolerated. For free-form discussion on no particular topic, the FreeBSD chat mailing list is freely available and should be used instead.
- No posting should be made to more than 2 mailing lists, and only to 2 when a clear and obvious need to post to both lists exists. For most lists, there is already a great deal of subscriber overlap and except for the most esoteric mixes (say "-stable & -scsi"), there really is no reason to post to more than one list at a time. If a message is received with multiple mailing lists on the `Cc` line, trim the `Cc` line before replying. *The person who replies is still responsible for cross-posting, no matter who the originator might have been.*
- Personal attacks and profanity (in the context of an argument) are not allowed, and that includes users and developers alike. Gross breaches of netiquette, like excerpting or reposting private mail when permission to do so was not and would not be forthcoming, are frowned upon but not specifically enforced.
- Advertising of non-FreeBSD related products or services is strictly prohibited and will result in an immediate ban if it is clear that the offender is advertising by spam.

C.2.3. Filtering on the Mailing Lists

The FreeBSD mailing lists are filtered in multiple ways to avoid the distribution of spam, viruses, and other unwanted emails. The filtering actions described in this section do not include all those used to protect the mailing lists.

Only certain types of attachments are allowed on the mailing lists. All attachments with a MIME content type not found in the list below will be stripped before an email is distributed on the mailing lists.

- application/octet-stream
- application/pdf
- application/pgp-signature
- application/x-pkcs7-signature
- message/rfc822
- multipart/alternative
- multipart/related
- multipart/signed
- text/html
- text/plain
- text/x-diff
- text/x-patch



Some of the mailing lists might allow attachments of other MIME content types, but the above list should be applicable for most of the mailing lists.

If a multi-part message includes text/plain and text/html parts:

- addressees will receive both parts
- lists.freebsd.org will present text/plain with an option to view original text (source, with raw HTML amongst the parts).

If text/plain does not accompany text/html:

- there will be conversion from HTML to plain text.

C.3. Usenet Newsgroups

In addition to two FreeBSD specific newsgroups, there are many others in which FreeBSD is discussed or are otherwise relevant to FreeBSD users.

C.3.1. BSD Specific Newsgroups

- comp.unix.bsd.freebsd.announce
- comp.unix.bsd.freebsd.misc
- de.comp.os.unix.bsd (German)
- fr.comp.os.bsd (French)

C.3.2. Other UNIX® Newsgroups of Interest

- comp.unix
- comp.unix.questions
- comp.unix.admin
- comp.unix.programmer
- comp.unix.shell
- comp.unix.misc
- comp.unix.bsd

C.3.3. X Window System

- comp.windows.x

Appendix D: OpenPGP Keys

The OpenPGP keys of the FreeBSD.org officers are shown here. These keys can be used to verify a signature or send encrypted email to one of the officers. A full list of FreeBSD OpenPGP keys is available in the PGP Keys article. The complete keyring can be downloaded at pgpkeyring.txt.

D.1. Officers

D.1.1. Security Officer Team <security-officer@FreeBSD.org>

```
pub   rsa4096/D9AD2A18057474CB 2022-12-11 [C] [expires: 2030-01-24]
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D.1.2. Core Team Secretary <core-secretary@FreeBSD.org>

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pub   rsa4096/4D632518C3546B05 2024-02-17 [SC] [expires: 2028-02-08]
      Key fingerprint = 1A23 6A92 528D 00DD 7965  76FE 4D63 2518 C354 6B05
uid                           FreeBSD Core Team Secretary <core-
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secretary@FreeBSD.org>
sub   rsa4096/CABFDE12CA516ED2 2024-02-17 [E] [expires: 2028-02-08]
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D.1.3. Ports Management Team Secretary <portmgr-secretary@FreeBSD.org>

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pub   ed25519/E3C401F60D709D59 2023-03-06 [SC] [expires: 2027-03-05]
      Key fingerprint = BED4 A1D3 6555 B681 2E9F  ABDA E3C4 01F6 0D70 9D59
uid                           FreeBSD Ports Management Team Secretary <portmgr-
secretary@FreeBSD.org>
sub   cv25519/2C92B55E27A641C3 2023-03-06 [E] [expires: 2027-03-05]
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D.1.4. doceng-secretary@FreeBSD.org

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pub   rsa2048/E1C03580AEB45E58 2019-10-31 [SC] [expires: 2022-10-30]
      Key fingerprint = F24D 7B32 B864 625E 5541  A0E4 E1C0 3580 AEB4 5E58
uid                           FreeBSD Doceng Team Secretary <doceng-
secretary@freebsd.org>
sub   rsa2048/9EA8D713509472FC 2019-10-31 [E] [expires: 2022-10-30]
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```

```
=MQKT
-----END PGP PUBLIC KEY BLOCK-----
```

FreeBSD Glossary

This glossary contains terms and acronyms used within the FreeBSD community and documentation.

A

ACL

See Access Control List.

ACPI

See Advanced Configuration and Power Interface.

AMD

See Automatic Mount Daemon.

AML

See ACPI Machine Language.

API

See Application Programming Interface.

APIC

See Advanced Programmable Interrupt Controller.

APM

See Advanced Power Management.

APOP

See Authenticated Post Office Protocol.

ASL

See ACPI Source Language.

ATA

See Advanced Technology Attachment.

ATM

See Asynchronous Transfer Mode.

ACPI Machine Language

Pseudocode, interpreted by a virtual machine within an ACPI-compliant operating system, providing a layer between the underlying hardware and the documented interface presented to the OS.

ACPI Source Language

The programming language AML is written in.

Access Control List

A list of permissions attached to an object, usually either a file or a network device.

Advanced Configuration and Power Interface

A specification which provides an abstraction of the interface the hardware presents to the operating system, so that the operating system should need to know nothing about the underlying hardware to make the most of it. ACPI evolves and supersedes the functionality provided previously by APM, PNPBIOS and other technologies, and provides facilities for controlling power consumption, machine suspension, device enabling and disabling, etc.

Application Programming Interface

A set of procedures, protocols and tools that specify the canonical interaction of one or more program parts; how, when and why they do work together, and what data they share or operate on.

Advanced Power Management

An API enabling the operating system to work in conjunction with the BIOS in order to achieve power management. APM has been superseded by the much more generic and powerful ACPI specification for most applications.

Advanced Programmable Interrupt Controller

Advanced Technology Attachment

Asynchronous Transfer Mode

Authenticated Post Office Protocol

Automatic Mount Daemon

A daemon that automatically mounts a filesystem when a file or directory within that filesystem is accessed.

B

BAR

See Base Address Register.

BIND

See Berkeley Internet Name Domain.

BIOS

See Basic Input/Output System.

BSD

See Berkeley Software Distribution.

Base Address Register

The registers that determine which address range a PCI device will respond to.

Basic Input/Output System

The definition of BIOS depends a bit on the context. Some people refer to it as the ROM chip with a basic set of routines to provide an interface between software and hardware. Others refer to it as the set of routines contained in the chip that help in bootstrapping the system. Some might also refer to it as the screen used to configure the bootstrapping process. The BIOS is PC-specific but other systems have something similar.

Berkeley Internet Name Domain

An implementation of the DNS protocols.

Berkeley Software Distribution

This is the name that the Computer Systems Research Group (CSRG) at The University of California at Berkeley gave to their improvements and modifications to AT&T's 32V UNIX®. FreeBSD is a descendant of the CSRG work.

Bikeshed Building

A phenomenon whereby many people will give an opinion on an uncomplicated topic, whilst a complex topic receives little or no discussion. See the FAQ for the origin of the term.

C

CD

See Carrier Detect.

CHAP

See Challenge Handshake Authentication Protocol.

CLIP

See Classical IP over ATM.

COFF

See Common Object File Format.

CPU

See Central Processing Unit.

CTS

See Clear To Send.

Carrier Detect

An RS232C signal indicating that a carrier has been detected.

Central Processing Unit

Also known as the processor. This is the brain of the computer where all calculations take place. There are a number of different architectures with different instruction sets. Among the more well-known are the Intel-x86 and derivatives, Arm, and PowerPC.

Challenge Handshake Authentication Protocol

A method of authenticating a user, based on a secret shared between client and server.

Classical IP over ATM

Clear To Send

An RS232C signal giving the remote system permission to send data.

See Also Request To Send.

Common Object File Format

D

DAC

See Discretionary Access Control.

DDB

See Debugger.

DES

See Data Encryption Standard.

DHCP

See Dynamic Host Configuration Protocol.

DNS

See Domain Name System.

DRM

See Direct Rendering Manager.

DSDT

See Differentiated System Description Table.

DSR

See Data Set Ready.

DTR

See Data Terminal Ready.

DVMRP

See Distance-Vector Multicast Routing Protocol.

Discretionary Access Control

Data Encryption Standard

A method of encrypting information, traditionally used as the method of encryption for UNIX® passwords and the crypt(3) function.

Data Set Ready

An RS232C signal sent from the modem to the computer or terminal indicating a readiness to send and receive data.

See Also Data Terminal Ready.

Data Terminal Ready

An RS232C signal sent from the computer or terminal to the modem indicating a readiness to send and receive data.

Debugger

An interactive in-kernel facility for examining the status of a system, often used after a system has crashed to establish the events surrounding the failure.

Differentiated System Description Table

An ACPI table, supplying basic configuration information about the base system.

Distance-Vector Multicast Routing Protocol

Domain Name System

The system that converts humanly readable hostnames (i.e., mail.example.net) to Internet addresses and vice versa.

Direct Rendering Manager

The drm(7) kernel module allows client applications direct access to the graphical hardware via the Direct Rendering Infrastructure.

Dynamic Host Configuration Protocol

A protocol that dynamically assigns IP addresses to a computer (host) when it requests one from the server. The address assignment is called a “lease”.

E

ECOFF

See Extended COFF.

ELF

See Executable and Linking Format.

ESP

See Encapsulated Security Payload.

Encapsulated Security Payload

Executable and Linking Format

Extended COFF

F

FADT

See Fixed ACPI Description Table.

FAT

See File Allocation Table.

FAT16

See File Allocation Table (16-bit).

FTP

See File Transfer Protocol.

File Allocation Table

File Allocation Table (16-bit)

File Transfer Protocol

A member of the family of high-level protocols implemented on top of TCP which can be used to transfer files over a TCP/IP network.

Fixed ACPI Description Table

G

GUI

See Graphical User Interface.

Giant

The name of a mutual exclusion mechanism (a sleep `mutex`) that protects a large set of kernel resources. Although a simple locking mechanism was adequate in the days where a machine might have only a few dozen processes, one networking card, and certainly only one processor, in current times it is an unacceptable performance bottleneck. FreeBSD developers are actively working to replace it with locks that protect individual resources, which will allow a much greater degree of parallelism for both single-processor and multi-processor machines.

Graphical User Interface

A system where the user and computer interact with graphics.

H

HTML

See HyperText Markup Language.

HUP

See HangUp.

HangUp

HyperText Markup Language

The markup language used to create web pages.

I

I/O

See Input/Output.

IASL

See Intel's ASL compiler.

IMAP

See Internet Message Access Protocol.

IP

See Internet Protocol.

IPFW

See IP Firewall.

IPP

See Internet Printing Protocol.

IPv4

See IP Version 4.

IPv6

See IP Version 6.

ISP

See Internet Service Provider.

IP Firewall

IP Version 4

The IP protocol version 4, which uses 32 bits for addressing. This version is still the most widely used, but it is slowly being replaced with IPv6.

See Also IP Version 6.

IP Version 6

The new IP protocol. Invented because the address space in IPv4 is running out. Uses 128 bits for addressing.

Input/Output

Intel's ASL compiler

Intel's compiler for converting ASL into AML.

Internet Message Access Protocol

A protocol for accessing email messages on a mail server, characterised by the messages usually being kept on the server as opposed to being downloaded to the mail reader client.

See Also Post Office Protocol Version 3.

Internet Printing Protocol

Internet Protocol

The packet transmitting protocol that is the basic protocol on the Internet. Originally developed at the U.S. Department of Defense and an extremely important part of the TCP/IP stack. Without the Internet Protocol, the Internet would not have become what it is today. For more information, see RFC 791.

Internet Service Provider

A company that provides access to the Internet.

K

KAME

Japanese for "turtle", the term KAME is used in computing circles to refer to the KAME Project, who work on an implementation of IPv6.

KDC

See Key Distribution Center.

KLD

See Kernel ld(1).

KMS

See Kernel Mode Setting.

KSE

See Kernel Scheduler Entities.

KVA

See Kernel Virtual Address.

Kbps

See Kilo Bits Per Second.

Setting the display mode in kernel space.

Kernel ld(1)

A method of dynamically loading functionality into a FreeBSD kernel without rebooting the system.

Kernel Scheduler Entities

A kernel-supported threading system. See the KSE Project home page for further details.

Kernel Virtual Address

Key Distribution Center

Kilo Bits Per Second

Used to measure bandwidth (how much data can pass a given point at a specified amount of time). Alternates to the Kilo prefix include Mega, Giga, Tera, and so forth.

L

LAN

See Local Area Network.

LOR

See Lock Order Reversal.

LPD

See Line Printer Daemon.

Line Printer Daemon

Local Area Network

A network used on a local area, e.g. office, home, or so forth.

Lock Order Reversal

The FreeBSD kernel uses a number of resource locks to arbitrate contention for those resources. A run-time lock diagnostic system found in FreeBSD-CURRENT kernels (but removed for releases), called witness(4), detects the potential for deadlocks due to locking errors. (witness(4) is actually slightly conservative, so it is possible to get false positives.) A true positive report indicates that "if you were unlucky, a deadlock would have happened here".

True positive LORs tend to get fixed quickly, so check https://lists.FreeBSD.org/subscription/freebsd-current and the LORs Seen page before posting to the mailing lists.

M

MAC

See Mandatory Access Control.

MADT

See Multiple APIC Description Table.

MFC

See Merge From Current.

MFH

See Merge From Head.

MFS

See Merge From Stable.

MFV

See Merge From Vendor.

MIT

See Massachusetts Institute of Technology.

MLS

See Multi-Level Security.

MOTD

See Message Of The Day.

MTA

See Mail Transfer Agent.

MUA

See Mail User Agent.

Mail Transfer Agent

An application used to transfer email. An MTA has traditionally been part of the BSD base system. Today Sendmail is included in the base system, but there are many other MTAs, such as postfix, qmail and Exim.

Mail User Agent

An application used by users to display and write email.

Mandatory Access Control

Massachusetts Institute of Technology

Merge From Current

To merge functionality or a patch from the -CURRENT branch to another, most often -STABLE.

Merge From Head

To merge functionality or a patch from a repository HEAD to an earlier branch.

Merge From Stable

In the normal course of FreeBSD development, a change will be committed to the -CURRENT branch for testing before being merged to -STABLE. On rare occasions, a change will go into -STABLE first and then be merged to -CURRENT.

This term is also used when a patch is merged from -STABLE to a security branch.

See Also Merge From Current.

Merge From Vendor

Message Of The Day

A message, usually shown on login, often used to distribute information to users of the system.

Multi-Level Security

Multiple APIC Description Table

N

NAT

See Network Address Translation.

NDISulator

See Project Evil.

NFS

See Network File System.

NTFS

See New Technology File System.

NTP

See Network Time Protocol.

Network Address Translation

A technique where IP packets are rewritten on the way through a gateway, enabling many machines behind the gateway to effectively share a single IP address.

Network File System

New Technology File System

A filesystem developed by Microsoft and available in its “New Technology” operating systems, such as Windows® 2000, Windows NT® and Windows® XP.

Network Time Protocol

A means of synchronizing clocks over a network.

O

OBE

See Overtaken By Events.

ODMR

See On-Demand Mail Relay.

OS

See Operating System.

On-Demand Mail Relay

Operating System

A set of programs, libraries and tools that provide access to the hardware resources of a computer. Operating systems range today from simplistic designs that support only one program running at a time, accessing only one device to fully multi-user, multi-tasking and multi-process systems that can serve thousands of users simultaneously, each of them running dozens of different applications.

Overtaken By Events

Indicates a suggested change (such as a Problem Report or a feature request) which is no longer relevant or applicable due to such things as later changes to FreeBSD, changes in networking standards, the affected hardware having since become obsolete, and so forth.

P

PAE

See Physical Address Extensions.

PAM

See Pluggable Authentication Modules.

PAP

See Password Authentication Protocol.

PC

See Personal Computer.

PCNSFD

See crossref:glossary[pcnfsd-glossary,Personal Computer Network File System Daemon].

PDF

See Portable Document Format.

PID

See Process ID.

POLA

See Principle Of Least Astonishment.

POP

See Post Office Protocol.

POP3

See Post Office Protocol Version 3.

PPD

See PostScript Printer Description.

PPP

See Point-to-Point Protocol.

PPPoA

See PPP over ATM.

PPPoE

See PPP over Ethernet.

PPP over ATM

PPP over Ethernet

PR

See Problem Report.

PXE

See Preboot eXecution Environment.

Password Authentication Protocol

Personal Computer

Personal Computer Network File System Daemon

Physical Address Extensions

A method of enabling access to up to 64 GB of RAM on systems which only physically have a 32-bit wide address space (and would therefore be limited to 4 GB without PAE).

Pluggable Authentication Modules

Point-to-Point Protocol

Pointy Hat

A mythical piece of headgear, much like a dunce cap, awarded to any FreeBSD committer who breaks the build, makes revision numbers go backwards, or creates any other kind of havoc in the source base. Any committer worth his or her salt will soon accumulate a large collection. The usage is (almost always?) humorous.

Portable Document Format

Post Office Protocol

See Also Post Office Protocol Version 3.

Post Office Protocol Version 3

A protocol for accessing email messages on a mail server, characterised by the messages usually

being downloaded from the server to the client, as opposed to remaining on the server.

See Also Internet Message Access Protocol.

PostScript Printer Description

Preboot eXecution Environment

Principle Of Least Astonishment

As FreeBSD evolves, changes visible to the user should be kept as unsurprising as possible. For example, arbitrarily rearranging system startup variables in /etc/defaults/rc.conf violates POLA. Developers consider POLA when contemplating user-visible system changes.

PRIME

A method of multiple physical graphics coprocessors coexisting by sharing their direct memory access buffers.

Problem Report

A description of some kind of problem that has been found in either the FreeBSD source or documentation. See Writing FreeBSD Problem Reports.

Process ID

A number, unique to a particular process on a system, which identifies it and allows actions to be taken against it.

Project Evil

The working title for the NDISulator, written by Bill Paul, who named it referring to how awful it is (from a philosophical standpoint) to need to have something like this in the first place. The NDISulator is a special compatibility module to allow Microsoft Windows™ NDIS miniport network drivers to be used with FreeBSD/i386. This is usually the only way to use cards where the driver is closed-source. See src/sys/compat/ndis/subr_ndis.c.

R

RA

See Router Advertisement.

RAID

See Redundant Array of Inexpensive Disks.

RAM

See Random Access Memory.

RD

See Received Data.

RFC

See Request For Comments.

RISC

See Reduced Instruction Set Computer.

RPC

See Remote Procedure Call.

RS232C

See Recommended Standard 232C.

RTS

See Request To Send.

Random Access Memory

Revision Control System

The *Revision Control System (RCS)* is one of the oldest software suites that implement "revision control" for plain files. It allows the storage, retrieval, archival, logging, identification and merging of multiple revisions for each file. RCS consists of many small tools that work together. It lacks some of the features found in more modern revision control systems, like Git, but it is very simple to install, configure, and start using for a small set of files.

See Also Subversion.

Received Data

An RS232C pin or wire that data is received on.

See Also Transmitted Data.

Recommended Standard 232C

A standard for communications between serial devices.

Reduced Instruction Set Computer

An approach to processor design where the operations the hardware can perform are simplified but made as general purpose as possible. This can lead to lower power consumption, fewer transistors and in some cases, better performance and increased code density. Examples of RISC processors include the Alpha, SPARC®, ARM® and PowerPC®.

Redundant Array of Inexpensive Disks

Remote Procedure Call

Request For Comments

A set of documents defining Internet standards, protocols, and so forth. See www.rfc-editor.org.

Also used as a general term when someone has a suggested change and wants feedback.

Request To Send

An RS232C signal requesting that the remote system commences transmission of data.

See Also Clear To Send.

Router Advertisement

S

SCI

See System Control Interrupt.

SCSI

See Small Computer System Interface.

SG

See Signal Ground.

SLAAC

See StateLess Address AutoConfiguration.

SMB

See Server Message Block.

SMP

See Symmetric MultiProcessor.

SMTP

See Simple Mail Transfer Protocol.

SMTP AUTH

See SMTP Authentication.

SSH

See Secure Shell.

STR

See Suspend To RAM.

SVN

See Subversion.

StateLess Address AutoConfiguration

SMTP Authentication

Server Message Block

Signal Ground
: An RS232 pin or wire that is the ground reference for the signal.

Simple Mail Transfer Protocol

Secure Shell

Small Computer System Interface

Subversion
: Subversion is a version control system currently used by the FreeBSD project.

Suspend To RAM

Symmetric MultiProcessor

System Control Interrupt

T

TCP
: See Transmission Control Protocol.

TCP/IP
: See crossref:glossary[tcpip-glossary,Transmission Control Protocol/Internet Protocol].

TD
: See Transmitted Data.

TFTP
: See Trivial FTP.

TGT

See Ticket-Granting Ticket.

TSC

See Time Stamp Counter.

Ticket-Granting Ticket

Time Stamp Counter

A profiling counter internal to modern Pentium® processors that counts core frequency clock ticks.

Transmission Control Protocol

A protocol that sits on top of (e.g.) the IP protocol and guarantees that packets are delivered in a reliable, ordered, fashion.

Transmission Control Protocol/Internet Protocol

The term for the combination of the TCP protocol running over the IP protocol. Much of the Internet runs over TCP/IP.

Transmitted Data

An RS232C pin or wire that data is transmitted on.

See Also Received Data.

Trivial FTP

U

UDP

See User Datagram Protocol.

UFS1

See Unix File System Version 1.

UFS2

See Unix File System Version 2.

UID

See User ID.

URL

See Uniform Resource Locator.

USB

See Universal Serial Bus.

Uniform Resource Locator

A method of locating a resource, such as a document on the Internet and a means to identify that resource.

Unix File System Version 1

The original UNIX® file system, sometimes called the Berkeley Fast File System.

Unix File System Version 2

An extension to UFS1, introduced in FreeBSD 5-CURRENT. UFS2 adds 64 bit block pointers (breaking the 1T barrier), support for extended file storage and other features.

Universal Serial Bus

A hardware standard used to connect a wide variety of computer peripherals to a universal interface.

User ID

A unique number assigned to each user of a computer, by which the resources and permissions assigned to that user can be identified.

User Datagram Protocol

A simple, unreliable datagram protocol which is used for exchanging data on a TCP/IP network. UDP does not provide error checking and correction like TCP.

V

VPN

See Virtual Private Network.

Virtual Private Network

A method of using a public telecommunication such as the Internet, to provide remote access to a localized network, such as a corporate LAN.

Colophon

This book is the combined work of hundreds of contributors to "The FreeBSD Documentation Project". The text is authored in AsciiDoc.